

Victoria Park Academy Shenzhen

CIE IGCSE Computer Science (0478) — YEAR 9 SEMESTER 1

WEEKS 7-11: DATA TRANSMISSION, HARDWARE, SOFTWARE

WEEK 7: DATA TRANSMISSION

LESSON 7.1 — PACKET SWITCHING

Topic: Data Transmission — Packet Switching

Duration: 40 minutes

Lesson Type: Theory with physical simulation

Syllabus Ref: 1.2.1

Learning Objective:

Explain packet switching; understand advantages over circuit switching.

Key Vocabulary:

- packet switching (分组交换 / fen zu jiao huan)
- data packet (数据包 / shu ju bao)
- header (报头 / bao tou)
- payload (有效载荷 / you xiao zai he)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "You need to send a 1000-page book to a friend in Beijing."
- | "Do you send one massive package or many small envelopes?"

3-12 min | DIRECT INSTRUCTION

- | Data split into packets. Each packet has: header (IP address, packet number, total),
- | payload (data), footer (error check). Packets take different routes, reassembled at destination.
- | Compare to circuit switching (old telephone: dedicated line).

12-25 min | GUIDED PRACTICE

- | Physical simulation: 5 students = "packets" with numbered cards.
- | Must travel from one side of room to other via different routes, reassemble in order.
- | One "packet" gets lost — what happens?

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 7A: Diagram packet structure. Label each part.
- | Explain: "Why is packet switching more reliable than circuit switching?"

35-40 min | PLENARY + HOMEWORK

- | "During Chinese New Year, millions send messages simultaneously."
- | "How does packet switching help?"
- | Homework: Write a letter to your grandmother explaining packet switching.
- | No technical jargon allowed.

EAL Support:

- Role-play essential — physical movement aids comprehension
- Bilingual: 分组交换 (packet switching), 数据包 (data packet)
- Sentence frame: "Packet switching is more reliable because..."

Differentiation:

Emerging | Pre-labelled diagram, peer support during simulation

Developing | Standard worksheet, self-check

Secure | Research: How does the internet handle packet loss?

Resources Needed:

- "Packet" cards (numbered) for simulation
- Worksheet 7A
- Large room space for movement

LESSON 7.2 — PACKET SWITCHING EXAM PRACTICE

Topic: Packet Switching — Simulation & Exam Questions

Duration: 40 minutes

Lesson Type: Simulation + exam practice

Syllabus Ref: 1.2.1

Learning Objective:

Describe packet switching in detail; answer exam questions; compare with circuit switching.

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | Draw a data packet. Label: header, payload, footer.
- | What information does each contain?

3-12 min | DIRECT INSTRUCTION

- | Review packet structure. Packet switching vs circuit switching:
- | advantages (efficiency, reliability, fault tolerance) and disadvantages (delay, out-of-order, complexity).

12-25 min | GUIDED PRACTICE

- | Worksheet 7B: Exam questions.
- | "Explain the advantages of packet switching over circuit switching." (4 marks)

25-35 min | INDEPENDENT PRACTICE

| Scenario: VPA network crashes. How does packet switching help recovery?

| Design a network diagram.

35-40 min | PLENARY + HOMEWORK

| "Why is circuit switching still used for some phone calls?"

| Homework: Research VoIP — does it use packet switching?

EAL Support:

- Diagram-based learning reduces language barrier
- Compare/contrast table for circuit vs packet switching

Differentiation:

Emerging | Sentence frames for comparison, pre-drawn diagrams

Developing | Standard exam questions, peer discussion

Secure | Design network topology, research modern protocols

Resources Needed:

- Worksheet 7B
- Blank paper for network diagrams
- Internet access for VoIP research

WORKSHEET 7A — PACKET SWITCHING

Name: _____ **Class:** _____ **Date:** _____

Learning Objective: Explain packet switching and understand its advantages

SECTION A: Packet Structure

Draw and label a data packet:

1. 1. What information is in the HEADER? _____

2. 2. What is the PAYLOAD? _____

3. 3. What is in the FOOTER? _____

SECTION B: Packet Switching vs Circuit Switching

| Feature | Packet Switching | Circuit Switching |

|-----|-----|-----|

| Connection | | |

| Route | | |

Reliability			
Efficiency			
Example			

SECTION C: Exam Questions

4. 4. Explain the advantages of packet switching over circuit switching. [4 marks]

5. 5. What happens if a packet is lost during transmission? [2 marks]

Bilingual Support:

packet switching = 分组交换 (fen zu jiao huan)

data packet = 数据包 (shu ju bao)

header = 报头 (bao tou)

payload = 有效载荷 (you xiao zai he)

WORKSHEET 7B — PACKET SWITCHING EXAM PRACTICE

Name: _____ Class: _____ Date: _____

SECTION A: Exam Questions

6. 1. Explain what is meant by packet switching. [2 marks]

7. 2. Describe two advantages of packet switching over circuit switching. [4 marks]

8. 3. A packet arrives at its destination out of order. Explain how this can happen and how it is resolved. [3 marks]

SECTION B: Scenario

9. 4. Victoria Park Academy Shenzhen has 500 students. During lunch, many students stream videos, send messages, and browse websites simultaneously.

(a) Why would circuit switching be inefficient for this situation? [2 marks]

(b) How does packet switching improve network performance? [2 marks]

ANSWER KEY — WEEK 7

Worksheet 7A:

- 10. 1. Header contains: source IP, destination IP, packet number, total packets
- 11. 2. Payload = the actual data being sent
- 12. 3. Footer contains: error checking data (e.g., checksum)
- 13. 4. Advantages: more efficient use of network, fault tolerant (if one route fails, packets can take another), no dedicated line needed, multiple users can share same line
- 14. 5. The destination requests retransmission; or the sender detects missing acknowledgment and resends

Worksheet 7B:

- 15. 1. Data is broken into small packets; each packet travels independently; reassembled at destination
- 16. 2. (1) More efficient — no dedicated line needed, bandwidth shared
(2) More reliable — if one route fails, packets can take another
- 17. 3. Packets take different routes; each packet has a sequence number; destination reassembles in correct order using sequence numbers
- 18. 4. (a) Circuit switching would need a dedicated line for each student — impossible with 500 users
(b) Packet switching shares bandwidth; packets from different users interleave; no wasted capacity when user is idle

WEEK 8: ERROR DETECTION & ENCRYPTION

LESSON 8.1 — ERROR DETECTION (PARITY & CHECKSUM)

Topic: Error Detection — Parity & Checksum

Duration: 40 minutes

Lesson Type: Theory with practice

Syllabus Ref: 1.2.2

Learning Objective:

Understand parity checks and checksums; explain basic encryption purpose.

Key Vocabulary:

- error detection (错误检测 / cuo wu jian ce)
- parity (奇偶校验 / ji ou jiao yan)
- checksum (校验和 / jiao yan he)
- encryption (加密 / jia mi)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "You receive a message: HELLO. But it says HELLO."
- | "How do you know it's wrong? How could you check?"

3-12 min | DIRECT INSTRUCTION

- | Parity (even/odd): add bit to make total 1s even.
- | Checksum: sum of data, transmitted with data, recalculated and compared.
- | Encryption: Caesar cipher demo (shift 3). Modern: HTTPS, SSL.

12-25 min | GUIDED PRACTICE

- | Add even parity to: 1011001.
- | Verify checksum for: [4, 7, 2, 9].
- | Decode Caesar cipher: "KHOOR" (shift 3).

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 8A: 5 parity problems, 3 checksums, 2 Caesar ciphers.
- | Extension: create your own cipher.

35-40 min | PLENARY + HOMEWORK

- | "Why is a checksum better than a parity check? When might both fail?"
- | Homework: Research: What does the "S" in HTTPS stand for?
- | Why does it matter when you use WeChat Pay?

EAL Support:

- Caesar cipher = physical wheel (paper template).
Parity = "counting marbles" — if odd, add one to make even
- Real-world: WeChat Pay security, why HTTPS matters

Differentiation:

Emerging | Physical parity wheel, calculator for checksums

Developing | Standard worksheet, peer checking

Secure | Create own cipher, research modern encryption

Resources Needed:

- Worksheet 8A

- Caesar cipher wheels (paper)
- Calculator for checksums

LESSON 8.2 — ENCRYPTION & MODERN SECURITY

Topic: Encryption — Modern Security

Duration: 40 minutes

Lesson Type: Theory with real-world context

Syllabus Ref: 1.2.2

Learning Objective:

Explain modern encryption (SSL/TLS); understand why encryption is essential for online security.

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "You enter your password on a website."
- | "How do you know it's safe? What could go wrong?"

3-12 min | DIRECT INSTRUCTION

- | HTTP = unencrypted, HTTPS = encrypted (SSL/TLS).
- | Symmetric vs asymmetric encryption. Public/private keys.
- | Why encryption matters: banking, messaging, personal data.

12-25 min | GUIDED PRACTICE

- | Worksheet 8B: Identify secure vs insecure websites.
- | Exam questions on encryption.

25-35 min | INDEPENDENT PRACTICE

- | Scenario: Design a secure messaging system for Victoria Park Academy Shenzhen.
- | What encryption would you use? Why?

35-40 min | PLENARY + HOMEWORK

- | "Why is end-to-end encryption controversial for governments?"
- | Homework: Check if your favourite websites use HTTPS.

EAL Support:

- Real examples: show a phishing email (sanitised). "Spot the fake" activity
- Bilingual: 网络钓鱼 (phishing), 恶意软件 (malware)

Differentiation:

Emerging | Focus on HTTPS vs HTTP concept, visual examples

Developing | Standard worksheet, exam questions

Secure | Research: How does WhatsApp encryption work?

Resources Needed:

- Worksheet 8B
- Example phishing emails (sanitised)
- Website list for HTTPS check

WORKSHEET 8A — ERROR DETECTION

Name: _____ Class: _____ Date: _____

Learning Objective: Understand parity checks, checksums, and basic encryption

SECTION A: Parity Checks

Add an EVEN parity bit to make the total number of 1s even:

19.1. 1011001_ → _____

20.2. 1110000_ → _____

21.3. 0001111_ → _____

22.4. 1100110_ → _____

23.5. 1010101_ → _____

SECTION B: Checksums

24.6. Data: [4, 7, 2, 9]

Checksum = $4 + 7 + 2 + 9 =$ _____

Transmitted data: [4, 7, 2, 9, _____]

Receiver recalculates: $4 + 7 + 2 + 9 =$ _____

Does it match? _____

25.7. Data: [5, 3, 8, 1]

Checksum = _____

26.8. Received: [5, 3, 8, 1, 18]

Recalculate: _____

Match? _____ What does this tell you? _____

SECTION C: Caesar Cipher

Shift 3 (A→D, B→E, C→F...)

27.9. Decode: KHOOR → _____

28.10. Decode: ZHOFRPH → _____

29.11. Encode: HELLO → _____

30.12. Encode: COMPUTER → _____

SECTION D: Challenge

31.13. Create your own Caesar cipher with a different shift. Write a secret message to a friend.

Shift: _____

Message: _____

Encrypted: _____

WORKSHEET 8B — ENCRYPTION & MODERN SECURITY

Name: _____ Class: _____ Date: _____

SECTION A: Secure vs Insecure

For each website, mark SECURE (HTTPS) or INSECURE (HTTP):

32. 1. www.bank-of-china.com _____

33. 2. http://free-games.net _____

34. 3. https://www.victoria-park-academy-shenzhen _____

35. 4. www.shopping-mall.cn (no HTTPS) _____

36. 5. Why is HTTPS essential for online banking? _____

SECTION B: Exam Questions

37. 6. Explain what is meant by encryption. [2 marks]

38. 7. Describe the difference between HTTP and HTTPS. [2 marks]

39. 8. Explain why encryption is important when using public Wi-Fi. [3 marks]

SECTION C: Scenario

40. 9. Design a secure messaging system for Victoria Park Academy Shenzhen students.

(a) What encryption method would you use? _____

(b) Why is this better than sending messages without encryption? _____

(c) What are the limitations of your system? _____

ANSWER KEY — WEEK 8

Worksheet 8A:

41. 1. 10110010 (4 ones → even, parity = 0)

42. 2. 11100001 (3 ones → odd, parity = 1)

43. 3. 00011110 (4 ones → even, parity = 0)

44. 4. 11001100 (4 ones → even, parity = 0)

45. 5. 10101010 (4 ones → even, parity = 0)

46.6. Checksum = 22. Transmitted: [4,7,2,9,22]. Receiver: 22. Match: Yes.

47.7. Checksum = 17

48.8. Recalculate: $5+3+8+1 = 17$. Match: Yes. Data is likely correct.

49.9. KHOOR → HELLO

50.10. ZHOFRPH → WELCOME

51.11. HELLO → KHOOR

52.12. COMPUTER → FRPSXWHU

Worksheet 8B:

53.1. SECURE 2. INSECURE 3. SECURE 4. INSECURE

54.5. HTTPS encrypts data; prevents hackers from stealing passwords/account details

55.6. Encryption scrambles data so only authorised parties can read it

56.7. HTTP is unencrypted; HTTPS uses SSL/TLS to encrypt data between browser and server

57.8. Public Wi-Fi is insecure; hackers can intercept data; encryption prevents them reading passwords/messages; protects personal information

58.9. (a) End-to-end encryption (e.g., similar to WhatsApp/Signal)

(b) Prevents teachers/administrators from reading private messages; protects student privacy

(c) Cannot recover lost passwords; harder to monitor for bullying

WEEK 9: CPU & FDE CYCLE

LESSON 9.1 — CPU: FETCH-DECODE-EXECUTE CYCLE

Topic: CPU — Fetch-Decode-Execute Cycle

Duration: 40 minutes

Lesson Type: Theory with physical demonstration

Syllabus Ref: 1.3.1

Learning Objective:

Describe the FDE cycle; identify registers (PC, ACC, MAR, MDR, CIR).

Key Vocabulary:

- CPU (中央处理器 / zhong yang chu li qi)
- fetch-decode-execute (取指-译码-执行 / qu zhi - yi ma - zhi xing)
- register (寄存器 / ji cun qi)
- program counter (程序计数器 / cheng xu ji shu qi)
- accumulator (累加器 / lei jia qi)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| "A chef follows a recipe. What are the steps? How is this like a CPU?"

3-12 min | DIRECT INSTRUCTION

- | CPU components: ALU, CU, registers.
- | FDE cycle: Fetch (address from PC → MAR → memory → MDR → CIR),
- | Decode (CU interprets), Execute (ALU performs). PC increments.
- | Registers defined with analogies.

12-25 min | GUIDED PRACTICE

- | Trace table: step through 3 instructions.
- | Track PC, MAR, MDR, CIR, ACC values.

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 9A: Complete trace table for 5 instructions.
- | Extension: "What happens if the PC doesn't increment?"

35-40 min | PLENARY + HOMEWORK

- | "A CPU has a clock speed of 3GHz. How many FDE cycles per second?"
- | "Why does this matter for gaming?"
- | Homework: Draw the FDE cycle as a comic strip with 4 panels.
- | Label each register.

EAL Support:

- Chef analogy throughout: PC = bookmark (next step), MAR = asking for ingredient, MDR = receiving it, CIR = reading recipe, ALU = mixing bowl
- Physical role-play: 5 students as registers
- Bilingual: 取指-译码-执行 (FDE cycle)

Differentiation:

Emerging | Pre-filled trace table, focus on concept not speed

Developing | Standard trace table, peer checking

Secure | Extension: "What happens with conditional instructions?"

Resources Needed:

- Register cards for role-play
- Worksheet 9A
- Trace table templates

LESSON 9.2 — FDE CYCLE EXAM PRACTICE

Topic: FDE Cycle — Trace Tables & Exam Questions

Duration: 40 minutes

Lesson Type: Exam practice

Syllabus Ref: 1.3.1

Learning Objective:

Complete trace tables accurately; answer exam questions on the FDE cycle.

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| Quick fire: What does PC stand for? MAR? MDR? CIR? ACC?

3-12 min | DIRECT INSTRUCTION

| Review FDE cycle with emphasis on register changes.

| Common errors: forgetting to increment PC, confusing MAR and MDR.

12-25 min | GUIDED PRACTICE

| Worksheet 9B: Exam questions.

| "Describe the fetch-decode-execute cycle." (4 marks) + trace table completion.

25-35 min | INDEPENDENT PRACTICE

| Timed practice: Complete trace table in 10 minutes.

| Self-mark with answer key.

35-40 min | PLENARY + HOMEWORK

| "Why does a faster clock speed not always mean better performance?"

| Homework: Research: What is pipelining? How does it improve CPU performance?

EAL Support:

- Trace table format always on board — consistent layout reduces confusion
- Sentence frame: "During the fetch stage, the PC contains ____ and the MAR is set to ____"

Differentiation:

Emerging | Pre-filled first 2 rows, focus on understanding

Developing | Standard trace table, timed practice

Secure | Research pipelining, superscalar architecture

Resources Needed:

- Worksheet 9B
- Timer for practice
- Mark scheme

WORKSHEET 9A — CPU & FDE CYCLE

Name: _____ Class: _____ Date: _____

Learning Objective: Describe the FDE cycle and identify CPU registers

SECTION A: Register Definitions

Match the register to its function:

| Register | Function |

|-----|-----|

| PC (Program Counter) | |

| MAR (Memory Address Register) | |

MDR (Memory Data Register)	
CIR (Current Instruction Register)	
ACC (Accumulator)	

SECTION B: Trace Table

Complete the trace table for the following program:

Address Instruction

00 LOAD 05
 01 ADD 06
 02 STORE 07
 03 HALT

Assume: Memory[05] = 10, Memory[06] = 20

Step	PC	MAR	MDR	CIR	ACC	Action
1	00					Fetch
2						Decode
3						Execute
4						Fetch
5						Decode
6						Execute

SECTION C: Exam Questions

59.1. Describe the fetch-decode-execute cycle. [4 marks]

60.2. What is the purpose of the program counter? [2 marks]

Bilingual Support:

CPU = 中央处理器 (zhong yang chu li qi)

fetch-decode-execute = 取指-译码-执行 (qu zhi - yi ma - zhi xing)

register = 寄存器 (ji cun qi)

program counter = 程序计数器 (cheng xu ji shu qi)

accumulator = 累加器 (lei jia qi)

WORKSHEET 9B — FDE CYCLE EXAM PRACTICE

Name: _____ Class: _____ Date: _____

SECTION A: Exam Questions

61. 1. Describe the role of the following registers in the fetch-decode-execute cycle: [4 marks]

(a) MAR _____

(b) MDR _____

(c) CIR _____

(d) ACC _____

62. 2. Explain why the program counter is incremented during the fetch stage. [2 marks]

63. 3. A CPU has a clock speed of 2.5 GHz. Calculate how many fetch-decode-execute cycles it can perform in one second. [2 marks]

SECTION B: Trace Table Challenge

Complete the trace table for this program (timed: 10 minutes):

Address Instruction

10 LOAD 20

11 SUB 21

12 STORE 22

13 HALT

Assume: Memory[20] = 50, Memory[21] = 15

| Step | PC | MAR | MDR | CIR | ACC | Action |

|-----|----|-----|-----|-----|-----|-----|

| 1 | 10 | | | | Fetch |

| 2 | | | | | Decode |

| 3 | | | | | Execute |

| 4 | | | | | Fetch |

| 5 | | | | | Decode |

| 6 | | | | | Execute |

ANSWER KEY — WEEK 9

Worksheet 9A:

SECTION A:

PC = holds address of next instruction

MAR = holds address of memory location to access

MDR = holds data/instruction being transferred to/from memory

CIR = holds current instruction being decoded

ACC = stores results of arithmetic/logic operations

SECTION B: Trace table should show PC incrementing, ACC loading 10, then 30 (10+20), then storing 30 to address 07

64. 1. Fetch: PC → MAR → Memory → MDR → CIR; PC increments.

Decode: CU interprets instruction.

Execute: ALU performs operation. Repeat.

65. 2. PC holds address of next instruction; incremented so next fetch gets the following instruction

Worksheet 9B:

66. 1. (a) MAR holds memory address being accessed

(b) MDR holds data being transferred to/from memory

(c) CIR holds instruction being decoded

(d) ACC stores results of calculations

67. 2. PC incremented so CPU knows where to fetch next instruction; prevents infinite loop on same instruction

68. 3. 2.5 GHz = 2,500,000,000 cycles per second

Trace table: ACC = 50, then 35 (50-15), then stored to 22

WEEK 10: MEMORY

LESSON 10.1 — MEMORY: RAM, ROM, CACHE

Topic: Memory — RAM, ROM, Cache

Duration: 40 minutes

Lesson Type: Theory with comparison

Syllabus Ref: 1.3.2

Learning Objective:

Compare RAM, ROM, cache; explain virtual memory; understand why cache improves performance.

Key Vocabulary:

- RAM (随机存取存储器 / sui ji cun qu cun chu qi)
- ROM (只读存储器 / zhi du cun chu qi)
- cache (高速缓存 / gao su huan cun)
- virtual memory (虚拟内存 / xu ni nei cun)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "Your desk has space for 5 books. Your bookshelf has 100."
- | "Your library has 10,000. How do you decide what to keep close?"

3-12 min | DIRECT INSTRUCTION

- | RAM: volatile, read/write, stores running programs.
- | ROM: non-volatile, read-only, stores BIOS/boot instructions.
- | Cache: small, fast, stores frequently accessed data.
- | Virtual memory: using HDD as RAM (slow).
- | Hierarchy: registers → cache → RAM → virtual memory

12-25 min | GUIDED PRACTICE

- | Sorting activity: characteristics of RAM, ROM, cache into table.
- | "Which is fastest? Largest? Most expensive per MB?"

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 10A: Scenario: "You have 4GB RAM and open 20 Chrome tabs.
- | What happens? How does cache help?"

35-40 min | PLENARY + HOMEWORK

- | "A new CPU has 32MB cache vs old CPU with 4MB cache. Same clock speed."
- | "Which is faster? Why?"
- | Homework: Check your phone: How much RAM? Storage? Research: does it have cache?

EAL Support:

- Desk/bookshelf/library analogy = physical demonstration
- Bilingual: 随机存取存储器 (RAM), 只读存储器 (ROM), 高速缓存 (cache)

Differentiation:

Emerging | Pre-completed comparison table, focus on key differences

Developing | Standard worksheet, self-check

Secure | Research: Why is L1 cache faster than L2?

Resources Needed:

- Worksheet 10A
- Comparison table template
- Physical props (books for analogy)

LESSON 10.2 — MEMORY HIERARCHY & VIRTUAL MEMORY

Topic: Memory Hierarchy & Virtual Memory

Duration: 40 minutes

Lesson Type: Theory + exam practice

Syllabus Ref: 1.3.2

Learning Objective:

Explain memory hierarchy; describe virtual memory; answer exam questions.

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | List memory types from fastest to slowest.
- | From smallest to largest. From most expensive to cheapest.

3-12 min | DIRECT INSTRUCTION

- | Memory hierarchy pyramid: registers (fastest, smallest) → cache → RAM → virtual memory (slowest, largest).
- | Virtual memory: OS moves less-used data to HDD. Thrashing: excessive swapping.

12-25 min | GUIDED PRACTICE

- | Worksheet 10B: Exam questions.
- | "Explain why adding more RAM can improve computer performance." (3 marks)

25-35 min | INDEPENDENT PRACTICE

- | Design task: "You have \$500 to upgrade a computer."
- | "Would you buy more RAM, larger cache, or faster storage? Justify."

35-40 min | PLENARY + HOMEWORK

- | "Why is virtual memory much slower than RAM?"
- | Homework: Research: What is SSD caching? How does it help?

EAL Support:

- Pyramid diagram on board — visual hierarchy
- Sentence frame: "Cache is faster than RAM because it is ____ and closer to ____"

Differentiation:

Emerging | Pre-drawn pyramid, label only

Developing | Standard worksheet, exam questions

Secure | Research SSD caching, hybrid drives

Resources Needed:

- Worksheet 10B
- Pyramid diagram template
- Budget calculator

WORKSHEET 10A — MEMORY: RAM, ROM, CACHE

Name: _____ **Class:** _____ **Date:** _____

Learning Objective: Compare RAM, ROM, and cache; understand virtual memory

SECTION A: Memory Comparison

Feature	RAM	ROM	Cache
----- ----- ----- -----	----- ----- ----- -----	----- ----- ----- -----	----- ----- ----- -----
Volatile? (loses data when power off)			
Read/Write?			
Speed			
Size			
Purpose			

SECTION B: Memory Hierarchy

69. 1. Draw the memory hierarchy pyramid (fastest at top, slowest at bottom):

70. 2. Why is cache faster than RAM? _____

71. 3. What is virtual memory? _____

72. 4. Why is virtual memory slower than RAM? _____

SECTION C: Scenario

73. 5. You have 4GB RAM and open 20 Chrome tabs, each using 200MB.

(a) Total memory needed = ____ MB = ____ GB

(b) What happens when RAM is full? _____

(c) How does adding more RAM help? _____

Bilingual Support:

RAM = 随机存取存储器 (sui ji cun qu cun chu qi)

ROM = 只读存储器 (zhi du cun chu qi)

cache = 高速缓存 (gao su huan cun)

virtual memory = 虚拟内存 (xu ni nei cun)

WORKSHEET 10B — MEMORY HIERARCHY & EXAM PRACTICE

Name: _____ Class: _____ Date: _____

SECTION A: Exam Questions

74. 1. Explain why adding more RAM can improve computer performance. [3 marks]

75. 2. Describe what is meant by virtual memory. [2 marks]

76. 3. Explain why cache memory improves CPU performance. [3 marks]

77. 4. Compare RAM and ROM. [4 marks]

SECTION B: Design Task

78. 5. You have \$500 to upgrade a computer. Choose ONE upgrade:

(a) More RAM (8GB → 16GB) (b) Larger cache (4MB → 8MB) (c) Faster storage (HDD → SSD)

Your choice: _____

Justification: _____

ANSWER KEY — WEEK 10

Worksheet 10A:

RAM: Volatile, Read/Write, Fast, Medium (GB), Stores running programs

ROM: Non-volatile, Read-only, Slow, Small (MB), Stores BIOS/boot instructions

Cache: Volatile, Read/Write, Very fast, Very small (KB/MB), Stores frequently accessed data

79. 2. Cache is faster because it's closer to CPU and uses faster memory technology

80. 3. Virtual memory = using hard disk space as RAM when RAM is full

81. 4. Virtual memory is slower because hard disk is much slower than RAM

82. 5. (a) 4000MB = 4GB (b) Computer uses virtual memory (hard disk), slows down significantly

(c) More RAM means less need for virtual memory, faster performance

Worksheet 10B:

83. 1. More RAM means more programs can run simultaneously; less need for virtual memory; faster multitasking
84. 2. Virtual memory uses hard disk space as RAM when physical RAM is full; allows more programs to run
85. 3. Cache stores frequently used data closer to CPU; reduces need to access slower RAM; speeds up repeated operations
86. 4. RAM is volatile/read-write/fast/expensive; ROM is non-volatile/read-only/slow/cheap; RAM stores running programs, ROM stores permanent instructions
87. 5. (Any valid justification accepted) More RAM: best for multitasking; Cache: best for CPU-intensive tasks; SSD: best for faster boot and file access

WEEK 11: STORAGE & I/O

LESSON 11.1 — STORAGE (MAGNETIC, OPTICAL, SOLID-STATE)

Topic: Storage — Magnetic, Optical, Solid-State

Duration: 40 minutes

Lesson Type: Theory with comparison

Syllabus Ref: 1.3.3

Learning Objective:

Compare magnetic, optical, solid-state storage; select appropriate devices for scenarios.

Key Vocabulary:

- magnetic storage (磁存储 / ci cun chu)
- optical storage (光存储 / guang cun chu)
- solid-state storage (固态存储 / gu tai cun chu)
- sensor (传感器 / chuan gan qi)
- actuator (执行器 / zhi xing qi)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "You need to store: (a) a 2-hour 4K movie, (b) your phone's OS, (c) a school essay."
- | "What do you use for each?"

3-12 min | DIRECT INSTRUCTION

- | Magnetic (HDD): cheap, large, slow, moving parts.
- | Optical (CD/DVD): portable, fragile, small capacity.
- | Solid-state (SSD/flash): fast, expensive, no moving parts.
- | Emerging: cloud, M.2 NVMe.
- | I/O devices: input (keyboard, sensor), output (monitor, printer), storage (hybrid)

12-25 min | GUIDED PRACTICE

- | Decision matrix: cost, speed, capacity, durability, portability.

| Fill in for HDD, SSD, USB, DVD, Cloud.

25-35 min | INDEPENDENT PRACTICE

| Worksheet 11A: "Victoria Park Academy Shenzhen needs storage for: (1) student records (10 years),

| (2) CCTV footage (30 days), (3) exam software (temporary). Recommend and justify."

35-40 min | PLENARY + HOMEWORK

| "Why are SSDs replacing HDDs in laptops but not in data centres (yet)?"

| Homework: Research: Price per GB for HDD vs SSD in 2024.

| Predict: Will SSDs replace HDDs completely by 2030?

EAL Support:

- Bring physical examples: old CD, USB stick, photo of HDD internals
- "Touch and compare" station
- Sentence frame: "I choose ____ because it is ____ which means ____"

Differentiation:

Emerging | Pre-completed decision matrix, focus on key differences

Developing | Standard worksheet, scenario guidance

Secure | Research emerging storage technologies

Resources Needed:

- Physical storage examples (CD, USB, HDD photo)
- Worksheet 11A
- Price comparison websites

LESSON 11.2 — I/O DEVICES, SENSORS & EMBEDDED SYSTEMS

Topic: I/O Devices, Sensors & Embedded Systems

Duration: 40 minutes

Lesson Type: Theory with design

Syllabus Ref: 1.3.3

Learning Objective:

Explain how sensors and actuators work; give examples of embedded systems.

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| "Your air conditioner knows the room temperature and turns on automatically."

| "What components make this possible?"

3-12 min | DIRECT INSTRUCTION

| Sensor → processor → actuator → output.

| Examples: temperature sensor → microcontroller → motor (fan).

| Embedded systems: dedicated function, not general purpose.

| Examples: washing machine, microwave, car ABS, smartwatch.

12-25 min | GUIDED PRACTICE

| Diagram: draw a smart home system with 3 sensors and 2 actuators.

| Label data flow.

25-35 min | INDEPENDENT PRACTICE

| Worksheet 11B: "Design an embedded system for Victoria Park Academy Shenzhen:"

| "automatic lights, temperature control, or security. Draw block diagram."

35-40 min | PLENARY + HOMEWORK

| "Why is a smartphone NOT an embedded system? What makes it different?"

| Homework: Find 5 embedded systems in your home.

| List sensor, processor, and actuator for each.

EAL Support:

- "Input-Process-Output" loop = physical loop with arms
- Embedded system = "specialist, not generalist"
- Real-world: smart home devices, Xiaomi ecosystem

Differentiation:

Emerging | Pre-drawn block diagram, label only

Developing | Standard worksheet, design guidance

Secure | Design complex system with multiple sensors

Resources Needed:

- Worksheet 11B
- Blank paper for diagrams
- Smart home device examples

WORKSHEET 11A — STORAGE DEVICES

Name: _____ **Class:** _____ **Date:** _____

Learning Objective: Compare magnetic, optical, and solid-state storage

SECTION A: Storage Comparison

| Feature | Magnetic (HDD) | Optical (CD/DVD) | Solid-State (SSD/Flash) |

|-----|-----|-----|-----|

| Speed | | | |

| Capacity | | | |

| Durability | | | |

| Portability | | | |

| Cost per GB | | | |

| Moving parts? | | | |

SECTION B: Decision Making

88. 1. Victoria Park Academy Shenzhen needs storage for:

(a) Student records (10 years) — Recommend: _____ Justify: _____

(b) CCTV footage (30 days) — Recommend: _____ Justify: _____

(c) Exam software (temporary) — Recommend: _____ Justify: _____

89. 2. Your personal photos and videos:

(a) Best for backup: _____ Why? _____

(b) Best for everyday use: _____ Why? _____

SECTION C: Exam Questions

90. 3. Explain why an SSD is more suitable than an HDD for a laptop computer. [3 marks]

91. 4. Describe two disadvantages of optical storage compared to solid-state storage. [2 marks]

WORKSHEET 11B — SENSORS, ACTUATORS & EMBEDDED SYSTEMS

Name: _____ Class: _____ Date: _____

SECTION A: Input-Process-Output

92. 1. Draw the input-process-output loop for a smart air conditioner:

93. 2. Identify the sensor, processor, and actuator in:

(a) Automatic door: Sensor = _____ Actuator = _____

(b) Car airbag: Sensor = _____ Actuator = _____

(c) Smart light: Sensor = _____ Actuator = _____

SECTION B: Embedded Systems

94. 3. What is an embedded system? _____

95.4. Why is a smartphone NOT an embedded system? _____

96.5. Give three examples of embedded systems in your home:

(a) _____ Sensor: _____ Actuator: _____

(b) _____ Sensor: _____ Actuator: _____

(c) _____ Sensor: _____ Actuator: _____

SECTION C: Design Task

97.6. Design an embedded system for Victoria Park Academy Shenzhen:

Choose: Automatic lights / Temperature control / Security system

Your choice: _____

Draw the block diagram (sensor → processor → actuator):

Explain how it works: _____

ANSWER KEY — WEEK 11

Worksheet 11A:

Magnetic: Slow, Large, Fragile (moving parts), Not portable, Cheap, Yes moving parts

Optical: Very slow, Small, Very fragile, Portable, Very cheap, No moving parts (in disc)

Solid-State: Fast, Medium, Durable, Very portable, Expensive, No moving parts

98.1. (a) Magnetic (HDD) — cheap, large capacity for long-term storage

(b) Magnetic (HDD) — large capacity, cost-effective for temporary storage

(c) Solid-State (SSD) — fast, reliable for software

99.2. (a) Cloud or external HDD — backup, large capacity

(b) SSD or phone storage — fast access, everyday use

100. 3. SSD has no moving parts (more durable, less heat), faster read/write speeds, uses less battery power

101. 4. Slower, smaller capacity, more fragile, easily scratched

Worksheet 11B:

102. 1. Temperature sensor → microcontroller → motor (fan/compressor)

103. 2. (a) Motion sensor, motor (b) Accelerometer, explosive charge (c) Light sensor, LED/bulb

104. 3. A computer system with a dedicated function, built into a larger device

105. 4. Smartphone is general-purpose (can run many different apps); embedded systems have one dedicated function

106. 5. (Any valid examples accepted)

107. 6. (Any valid design with correct components accepted)

END OF FILE 3: WEEKS 7-11